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④

- $C_1 + \frac{C_2}{1+r} = C_1 + \frac{C_2}{1+r}$ Budget constraint.
- $NPV = \sum_{t=0}^n \frac{C_t}{(1+r)^t} \rightarrow \text{choose } NPV > 0$
- Payback Period = # years to recover.
(no time value money)
- Discounted Payback Period = PP w/ time value money.
- $IRR = \bar{r}$ s.t. $NPV = 0$
 - ↳ Cons: ¹ multiple IRR
 - ² Investment / Financing
 - ³ Scaling Problem

⑤

- Modified IRR
 - ↳ bring $\ominus$ CFs to $t=0$
 - & bring $\oplus$ CFs to final t . w/ (r)
 - then calculate IRR
- Scaling $\Rightarrow (B-A)$ Incremental IRR
 - $< r \Rightarrow$ small project
 - $> r \Rightarrow$ big project
- Profitability Index = $\frac{\sum_{t=1}^n \frac{C_t}{(1+r)^t}}{-C_0}$
 - ↳ accept $PI > 1$
- Equivalent Annual Cost : $NPV = X A \pi$

- Capital Budgeting Rules.

1. Ignore sunk cost.

2. Include opportunity cost.

3. Positive externalities $\Rightarrow$ synergy other $\rightarrow$ current

Negative externalities $\Rightarrow$ erosion current $\rightarrow$ other

4. $\bar{i}$ = nominal, r = real, π = Inflation

$$\underline{1+r = \frac{1+\bar{i}}{1+\pi}}$$

- $OCF = EBIT - \text{Taxes} + \text{Depreciation}$

$EBIT = \text{Revenue} - \text{Cost} - \text{Depreciation}$

$\text{Tax} = \text{Tax Rate} \times EBIT$

$$OCF = R - C - (T_c \times EBIT)$$

$$\underbrace{(R_t - C_t)(1 - T_c)}_{\text{ATOCF}} + \underbrace{T_c D}_{\text{CCATS}} = C_t (\text{operation})$$

after tax operation CF

Capital Cost allowance

Tax Shield

$$\Rightarrow NPV = C_0 + PVATOCF + PVCCATS$$

- Real & Nominal CF.

$$\hookrightarrow \text{Real} = \text{Nominal } CF_t \cdot \frac{1}{(1+\pi)^t}$$

$$\text{Nominal} = \text{Real } CF_t \cdot (1+\pi)^t$$

$\Rightarrow$ Same NPV.

(equipment)

- Half-Year Rule: buy half of machine this year, other half next year. (only depreciate half)

$$- PVCCATS = \frac{C \cdot d \cdot T_c}{r+d} \times \frac{1+0.5r}{1+r} - \frac{S \cdot d \cdot T_c}{r+d} \times \frac{1}{(1+r)^n}$$

C = Cost of asset, d = depreciation rate

n = lifetime of asset, S = salvage value

- NPV Uncertainties

→ Decision free $\Rightarrow$ Compare NPV of each option.

→ Scenario analysis $\Rightarrow$ calculate NPV of each scenario
($E(X)$) and combine NPV base on probability

→ Sensitivity analysis $\Rightarrow$ NPV sensitivity (distribution)

→ Break-even analysis $\Rightarrow$ σ^2 that make NPV = 0

$$\text{after-tax cost} = \frac{EAC + FC(1-T_c) - DT_c}{\text{contribution margin}}$$

$$\text{contribution margin} = (P - VC) \times (1 - T_c)$$

Break-Even point $\leftarrow$

- Expected Utility: $X \geq Y \Rightarrow E u(X) \geq E u(Y)$

$$\begin{cases} \text{Risk averse} & u(E(X)) > E(u(X)) & \text{or } u'' < 0 \\ \text{Risk neutral} & \dots = \dots & \text{or } u'' = 0 \\ \text{Risk loving} & \dots < \dots & \text{or } u'' > 0 \end{cases}$$

- Degree of Risk aversion:

$$\rightarrow \text{Absolute Risk Aversion coeff} = ARA = \frac{-u''(x)}{u'(x)}$$

$$\rightarrow \text{Relative coeff of Risk Aversion} = RRA = x \cdot ARA$$

- Certainty Equivalent, $CE(X) = \$$ amount for equal utility

- Mean-Variance Utility Function: $u(\mu, \sigma) = \mu - \frac{A}{2} \sigma^2$
 $\hookrightarrow A = \text{coeff of risk-aversion}$ $\downarrow$ $E(X)$ $\downarrow$ $\text{Var}(X)$

$$\hat{R} = \frac{1}{n} \sum r_t \quad \text{return at time } t$$

$$r_t = \frac{(P_{t+1} + D_t) - P_t}{P_t} = \frac{P_{t+1} - P_t}{P_t} + \frac{D_t}{P_t}$$

(Price yield) + (Dividend yield)

$$\sigma^2 = \frac{1}{n-1} \sum (R_i - \mu)^2 \quad (\text{volatility})$$

- Value At Risk = (VaR) $q_\alpha = \mu + \sigma Z_\alpha$ ★
 $\text{VaR}(X) = P(X \leq x) = \alpha$ (survival fcn of X)

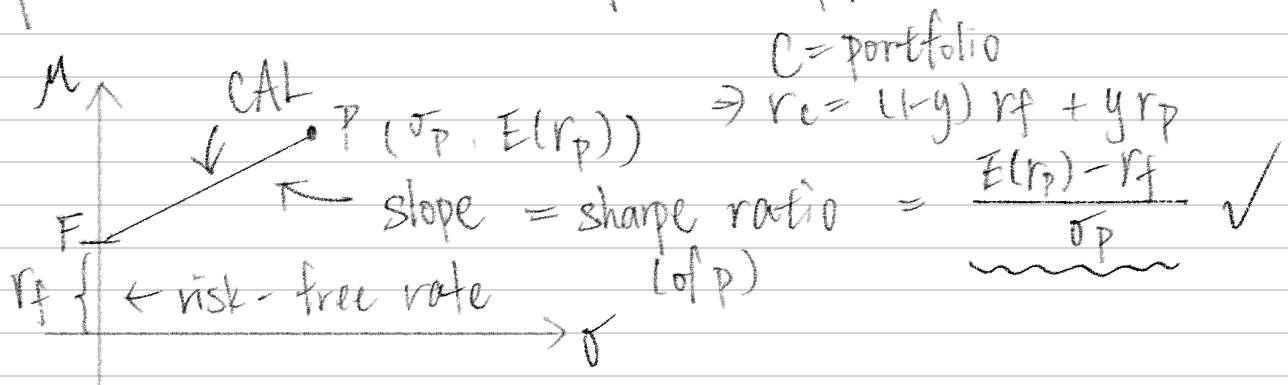
- Expected Loss:

$EL(X) = \text{average return, conditional } X \leq \text{VaR}(X)$

- Downside semi-Variance: $= \frac{1}{n-1} \sum_{r_i \leq \mu} (r_i - \mu)^2$

- Capital Allocation Line (r_f and r_p)

risky
port.
w/
 r_f asset.



$$\begin{cases} \mu_c = (1-y)r_f + yE(r_p) = r_f + y[E(r_p) - r_f] \\ \sigma_c^2 = y^2 \sigma_p^2 \Rightarrow y = \frac{\sigma_c}{\sigma_p} \end{cases}$$

$$\hookrightarrow \mu_c = r_f + \frac{[E(r_p) - r_f]}{\sigma_p} \sigma_c \checkmark \rightarrow \text{CAL}$$

↓ $\left(\mu_c - \frac{A}{2} \sigma_c^2\right)$
 plug-in result to mean-var utility function

$$\Rightarrow [r_f + y [E(r_p) - r_f]] - \frac{A}{2} \cdot y^2 \sigma_p^2$$

$$\left(\frac{d}{dy}\right) (\dots) = 0 \Rightarrow (E(r_p) - r_f) - \frac{A}{2} \cdot 2y^* \sigma_p^2 = 0$$

$$\checkmark \Rightarrow \underline{y^* = \frac{E(r_p) - r_f}{A \sigma_p^2}} \quad \text{optimal weight on risky}$$

- $w = \{w_1, w_2, \dots, w_n\}$ & $\sum w_i = 1$.

$$w_i = \frac{n_i P_i}{\sum_j n_j P_j} \quad \begin{array}{l} n = \# \text{ shares} \\ P = \text{price} \end{array}$$

$\oplus w \Rightarrow \text{long} \quad \ominus w \Rightarrow \text{Short (borrow \# asset)}$

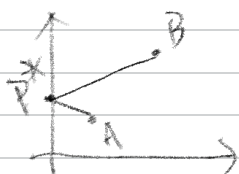
↳ Weight A and B ($\geq$ weight) =

$$\rho = \text{Corr}(R_A, R_B)$$

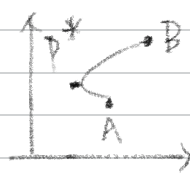
$$\star \begin{cases} \mu_c = w_A \mu_A + w_B \mu_B \\ \sigma_c^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2\rho w_A w_B \sigma_A \sigma_B \\ \quad = \underline{\bar{w}^T \Sigma \bar{w}} \end{cases}$$

$\rho = -1$ and let $\sigma_c = 0 \Rightarrow$ perfect hedge portfolio.

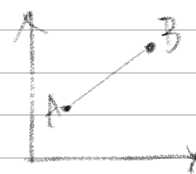
$\rho = -1$



$\rho = 0.3$



$\rho = 1$



Different weight to achieve P^* : minimum Var port.
 (left-most point)

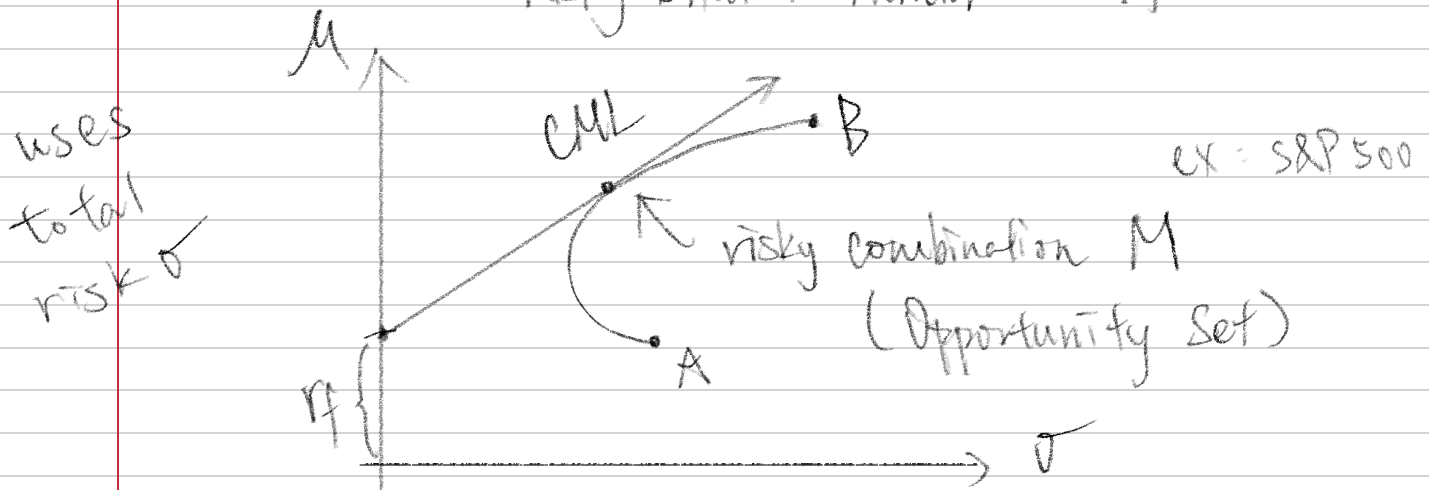
$$\checkmark \quad \underline{w_A^* = \frac{\sigma_B^2 - \rho \sigma_A \sigma_B}{\sigma_A^2 - 2\rho \sigma_A \sigma_B + \sigma_B^2}}$$

$$\underline{w_B^* = 1 - w_A^*}$$

(one risk-free asset)

- CML: draw CAL from r_f through risk portfolio given risk assets and risk-free assets.

↳ Risky Efficient Frontier ↳ r_f



$$\checkmark \quad \mu = r_f + \frac{E(r_M) - r_f}{\sigma_M} \sigma_C$$

↳ a variation of CAL

Given market combination (M.) market.

- N risky Asset:

$$\bar{w}^T = (w_1, w_2, \dots, w_n) \quad \bar{\mu}^T = (\mu_1, \mu_2, \dots, \mu_n)$$

$$\Sigma = \begin{pmatrix} \sigma_{11} & \dots & \sigma_{1n} \\ \sigma_{21} & \dots & \sigma_{2n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \dots & \sigma_{nn} \end{pmatrix}$$

$$\Rightarrow \underline{\mu_C} = \bar{w}^T \bar{\mu} = w_1 \mu_1 + w_2 \mu_2 + \dots$$

$$\underline{\sigma_C^2} = \bar{w}^T \Sigma \bar{w}$$

$$\underline{\bar{w}^*} = \frac{\Sigma^{-1} \bar{p}}{\bar{p}^T \Sigma^{-1} \bar{p}}$$

$$\beta = \gamma \Rightarrow \text{CML}$$



↓ $^+(\alpha)$ Jensen's.

use
systematic
risk β_i

- CAPM: $M_i = r_f + \beta_i [E(r_M) - r_f]$

↳ $\beta = \frac{\text{Cor}(r_i, r_M)}{\sigma_M^2}$

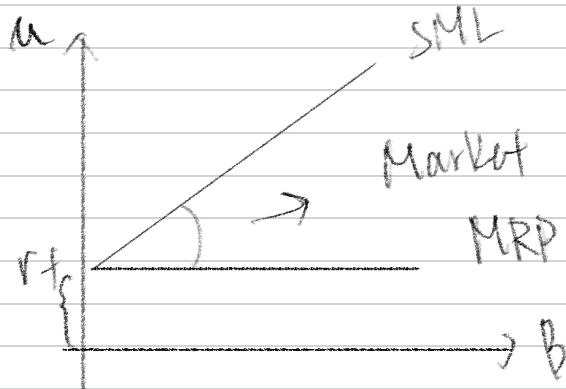
- SML: Gives expected return given β_i

β_i = sensitivity of asset i to changes in market.

- Treynor Ratio = $\frac{M_i - r_f}{\beta_i}$ *

Jensen's α

= actual return
- SML return.



- Arbitrage Pricing Theory

→ k common factors, $k=1 \Rightarrow$ Market Model

$\Rightarrow \underline{R_{it}} = \mu_i + \underbrace{m_i}_{\text{noise (mean=0)}} + \epsilon_i$ (A1)

actual return = expected return

+ unexpected return (due to risk)

and Risk = systematic + idiosyncratic

$\Rightarrow M_i = \beta_{i1} F_1 + \beta_{i2} F_2 + \dots + \beta_{ik} F_k$

F_k = common factors

β_{ij} = sensitivity of the i^{th} firm to the j^{th} factor.

ϵ_i = individual factor

(A2) if $\bar{i}$ is a well diversified portfolio, $\varepsilon_{\bar{i}} = 0$
 $\Rightarrow R_{\bar{i}} = M + m_{\bar{i}}$

(A3) two asset with identical payoff must have same price. (no-arbitrage)

γ = systematic risk (factor) left after assumption of "no-arbitrage" and "diversifiable idiosyncratic risk"

aka cross-sectional APT

↓

Theorem: $M_{\bar{i}} = r_f + \beta_{\bar{i}1}\gamma_1 + \beta_{\bar{i}2}\gamma_2 + \dots + \beta_{\bar{i}K}\gamma_K$

$K=1 \Rightarrow \underline{M_{\bar{i}}} = r_f + \beta_{\bar{i}} \cdot \gamma$

↳ Market port: $M_{\bar{i}} = r_f + \beta_{\bar{i}} \cdot M$

for $\bar{i}=M \Rightarrow M_M = r_f + \beta_M \cdot M \quad (\beta_M = 1)$

$\Rightarrow M_M = r_f + M$

$\Rightarrow M = M_M - r_f$

- Factor Effecting Betas

1- Cyclicity of Revenue

asset more sensitive to business cycle $\rightarrow$ high β

→ Operating leverage

$O.L. = \frac{\Delta\% \text{ Earnings}}{\Delta\% \text{ sales}}$ *

higher O.L. $\rightarrow$ high β

- Information asymmetry

① Agency problems: unequal knowledge b/w agent and principals.

② liquidity: (how fast trade is)

higher liquidity $\Rightarrow$ easier to trade $\Rightarrow$ lower cost of capital

③ Bid-Ask spread: buying price - selling price
 $\hookrightarrow$ higher if there is information asymmetry.

④ Financial leverage: previously, only equity

$$\Rightarrow SML: r_E = r_f + \beta_E [r_M - r_f]$$

Equity: "ownership contract", residual claims (dividend)
limited liability, (can't lose more than stock value)

Debt: "loan contract", interest is tax deductible.
failure to pay $\Rightarrow$ bankrupt.

levered firm: $D > 0$

unlevered firm: $D = 0$

- WACC (weighted average cost of capital)

$\hookrightarrow$ right interest (required return) rate for D/E

$$r_{WACC} = \frac{E}{D+E} r_E + \frac{D}{D+E} r_D (1 - T_c) \quad \rightarrow \text{if } \exists \text{ tax}$$

$\hookrightarrow$ from SML $\hookrightarrow$ loan interest.

$\uparrow$ liquidity $\Rightarrow \downarrow$ WACC

- EMH Effective Market hypothesis.

types of EMH:

1. Weak - price movement of a stock tnr reflects all info from price tdy. indep of past prices.

$$P_{t+1} = P_t + \varepsilon_t \quad (+g) \text{ with drift.}$$

ε_t error term indep. of P_t .

Random Walk Hypothesis.

ex: Gambler's Ruin:

A & B $\rightarrow$ \$N total

A has \$i, B has \$(N-i)

$P(A) = p$, $P(B) = 1-p$ (1\$ each game)

Game over when one's bankrupt.

$P(i)$ = Prob of A win if start with \$i.

$$P(i) = p - p(1+i) + (1-p)p(i-1)$$

$$P(0) = 0$$

$$P(N) = 1$$

} boundaries conditions.

$$\text{if } p = \frac{1}{2} \Rightarrow P(i) = i/N$$

Technical analysis = claim that they can see pattern in stock.

Weak form $\Rightarrow$ tech analysis are "guessing" (false)

2. Semi-Strong = price of stock take into account of all relevant public info.

⇒ you cannot "beat" the market.
w/ public information

3. Strong = you cannot "beat" the market w/
public & private info

((weak) Semi-strong) strong)

↳ violation of weak ⇒ violation of semi-strong
⇒ ... of strong

- Implication of EMH

1. Technical analysis ⇒ claim to "predict" future P
weak EMH true → they are just guessing

2. Fundamental analysis ⇒ analyze firm to check P
semi-strong EMH true → fundamental analysis guessing

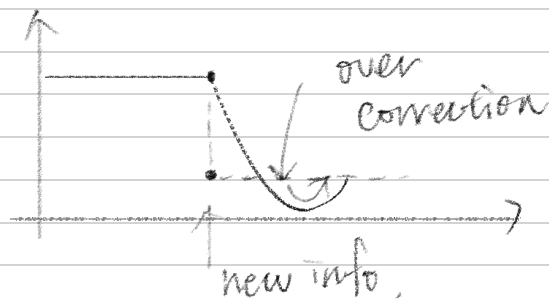
↳ Strong EMH true → private / insider information
is useless.

- Evidence for / against EMH Yes.

• test weak EMH → if we can fit Random Walk.

• " semi-strong " → if investment manager consistently
beating market ⇒ semi-strong violated.

p.s. EMH assumes price jump



Finance $\left\{ \begin{array}{l} \text{Behavioral} = \text{not rational} \\ \quad \quad \quad (\text{animal spirit.}) \\ \text{Classical} = \text{assume rationality} \\ \quad \quad \quad (\text{Dutch book theorem.}) \end{array} \right.$

- Capital Structure.

Q: does leverage effect value of firm?
optimal D/E ratio?

→ Modigliani-Miller thm.

Equity = Ownership Contract

Residual Claims, (return debt → priority)

Limit liability, (worst case stock → 0)

Market Cap = TPS × # of share.

Common Share → 1 share = 1 vote.

Superior share → 1 share > 1 vote.

(Limited Share → 1 share < 1 vote.

→ Equity vote for B.O.D and board select manager.

- Voting = Non-Cumulative $\Rightarrow$ board select sequentially, majority shareholder could select all.
Cumulative $\Rightarrow$ election run simultaneously, minority shareholder could play.

- Preferred Share = specifies amount of dividend / period.
b/w equity & debt.
 $\downarrow$ $\downarrow$
not pay dividend pre-specified payments.

$\hookrightarrow$ Fixed P-S = fix rate dividend.

Floating P-S = change with interest rate.

- Callable Share = Issuer have right to call / buy back at a pre-specific price.

Non-callable price > callable share.

- Retractable share = buyer have right to sell stock back to Issuer at predetermined price.

- Redeemable share = buyer has right to redeem the preferred share in exchange for common share.

- Reset Preferred: dividend rate changes. (/5 yr)

- Debt = loan contract

Interest paid is tax-deductible.

Failure to pay $\rightarrow$ bankrupt

{ Secure Debt = backed by asset (mortgage)
Unsecured Debt = backed by trust (debentures)

Types = long / short term, trade payable

• Indenture = contract b/w borrow / lender.

• Subordinated Debt: your debt is behind the

order. $\hookrightarrow$ "line" in the event of bankruptcy.
Senior / Junior debt holder $\rightarrow$ debentures
 $\rightarrow$ trade payable $\rightarrow$ shareholders.

- D/E ratio to maximize V ? ($= \frac{C}{r_{wacc}}$)
 $\Leftrightarrow r_{wacc}$ minimized. $\nwarrow$ value of firm.

Case ① "w/o tax"

• $MMI = \underline{V_{levered} = V_{unlevered}}$

$\Rightarrow$ D/E does not matter

ex: $V = 8000$, 400 shares

$\Rightarrow$ PPS = 20 $r_D = 10\%$

Unlevered = $D = 0$, $E = 8000$

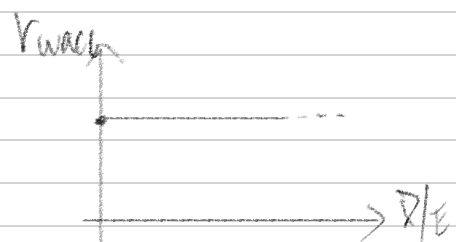
levered = $D = 4000$, $E = 4000$



200

200

400 share.
(buy back 200
Shares)



Example page.

Net Income

Unlevered:

		Recession	Normal	Expansion	
= Earnings:		400	1200	2000	
R.O.A.		5%	15%	25%	
R.O.E		5%	15%	25%	
⇒ EPS		1\$	3\$	5\$	400 share

Levered:

Earnings	400	1200	2000	
Interest	400	400	400	($r_D = 10\%$)
Net Income	0	800	1600	
⇒ EPS	0\$	4\$	8\$	200 share

(*) Suppose you want Invest 2000 in levered (50%)
 levered (EIX) 0\$ 400\$ 800\$

If its unlevered, strategy is to borrow 2000 and Invest 4000 in total ($r_D = 10\%$)

unlevered (EIX)	200	600	1000	
(r_D) Interest	-200	-200	-200	
	0\$	400\$	800\$	same

⇒ You could generate same payoff.

(*) Suppose you want Invest 2000 in unlevered (25%)
 unlevered (EIX) 100\$ 400\$ 500\$

If levered, Invest 1000 in and 1000 at $r = 10\%$

levered (EIX)	0	200	400	
Interest	+100	+100	+100	
	100\$	300\$	500\$	Same

Ex. Conclusion = firm's Value unaffected by D/E
 $\Rightarrow$ "Homemade leverage"

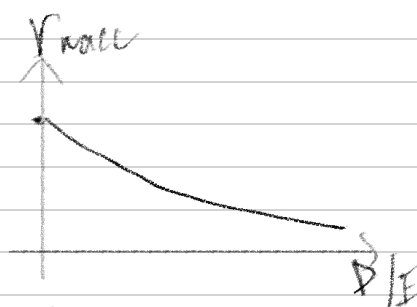
(Creating your own D/E ratio the same as the firm's)

• MMII : $r_E = r_u + D/E (r_u - r_D)$

higher $D/E \Rightarrow$ higher r_E

Case (2)

$\rightarrow (= \frac{E}{V} r_E + \frac{D}{V} r_D)$



• MMI : $V_L = V_u + D T_c$

(D rise $\Rightarrow$ r_{wacc} fall $\Rightarrow$ V increase

"optimum" is 100% debt)

Interest are tax deductible.

$\hookrightarrow$ def : Taxable Income = $EBIT - \text{Interest}$.

$$\text{Taxes} = T_c \times (EBIT - \text{Interest}).$$

$$\text{Taxes saved} = T_c \times \text{Interest}.$$

Levered firm =

• Bondholder's payoff : $r_D \cdot D$

• Shareholder's payoff = $(EBIT - r_D D) (1 - T_c)$

$$\text{total} = (EBIT - r_D D) (1 - T_c) + r_D D$$

$$= EBIT (1 - T_c) + r_D D T_c$$

$$V_L = \frac{EBIT (1 - T_c)}{r_u} + \frac{r_D D T_c}{r_D} = V_u + D \cdot T_c$$

$\Rightarrow$ • MMII : $r_E = r_u + \frac{D}{E} (1 - T_c) (r_u - r_D)$

Why D/r_D in r_w equation?

→ only used to express r_w from a levered firm you observed.

- Limits to Debt.

• Risk of bankruptcy (included in stock price)

↳ More debt → equity risk ↑ → value ↓

- Direct Cost

↳ lawyer's fee ex- 2009 Lehmann Brothers 100 mil
lost of reputation.

- Agency Cost

↳ Conflict of interest b/w management & debtholder

Why? Equity worst case is 0

⇒ so equity only cares abt upside (risky)
& debtholder prefer safer payoffs.

- Free Cash Flow Hypothesis (FCF)

• Firm generates more cash than it need for positive NPV projects, "free" cash flow are "wasted"

• These "wasted" CF could use to pay debt interest
⇒ Not "wasted"

- Signaling Theory: debt ⇒ healthy company

★ Summary: r_{wacc} ⇒ normal assets

r_E ⇒ fully financed w/ E, r_D ⇒ ... w/ debt.

- Calculate NPV for levered firms.

• WACC = find $r_{wacc} \Rightarrow$ use C_c and r_{wacc} .

• APV = Adjusted PV, $NPV_{\text{levered}} = NPV_{\text{unlevered}} + D T_c$

$\hookrightarrow$ formula assumes debt is perpetuity

$\rightarrow D T_c = \text{PV of tax break b/c of debt}$

b/c Interest tax shield = $T_c \cdot \text{Interest paid}$
 $= T_c \cdot (D r_D)$

Assume debt perpetuity $\Rightarrow \frac{T_c D r_D}{r_D} = T_c D \#$

P.S. $NPV_{\text{unlevered}} = \frac{EBIT(1-T_c)}{r_u}$

- Dividends = periodic pmts to shareholders.

• Cash dividend = $DPS \cdot \# \text{ share}$ (dividend per share)

• Stock dividend =

Percentage = 20% $\Rightarrow 100 \rightarrow 120$ shares

Stock Split = 2:1 $\Rightarrow 50 \rightarrow 100$ shares

Reverse stock split = 1:4 $\Rightarrow 100 \rightarrow 25$ shares.

Stock Pre-purchase = Firm buy off shares.

$\Rightarrow$ value of remaining share $\uparrow$

$\hookrightarrow$ Important dates:

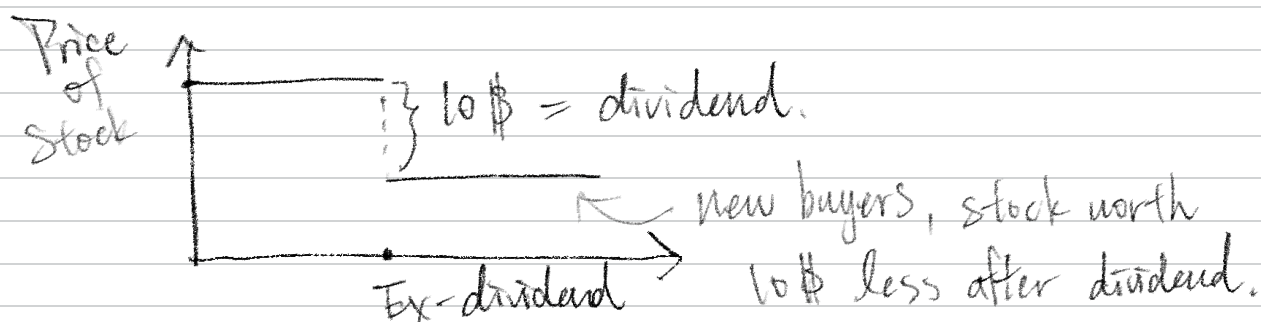
• Declaration date = BOD announces pmt

• Ex-dividend date = seller sells on or after this date keeps the dividend (regardless record)

• Record date = record who's entitled to dividend

(after Ex-dividend date)

- Payment date: money paid



$PPS = \text{Value of firm} / \text{share}$

- Dividend Policy Indifferent theorem =
 - According to MM, (ideal situation, $\times$ tax ...)
dividend policies are irrelevant. (to shareholders)
- Dividend policy matters in reality:
 - taxes: firms pays no tax on other firms dividends.
 - Clientele effect: diff client preference on dividends.
 - FCF Hyp = more dividend $\Rightarrow$ less free cash $\Rightarrow$ less waste $\Rightarrow$ higher Value.
 - Signaling = regular dividend $\Rightarrow$ company's health.
- Options:
 - derivative = a contract whose value is derived from (depends on) some underlying stock, index, bond
 - Call Option = gives buy right, not obligation, to buy an asset at a predetermined price on / before a predetermined date. $\searrow$

Underlying asset = stock

S_0 = price of stock today.

T = expiration date.

S_T = price of stock on expiration date.

X = pre-agreed price = strike price.

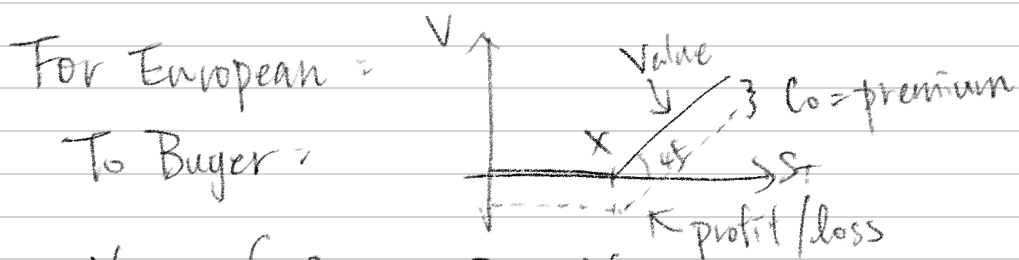
Buyer $\rightarrow$ Holder, long position.

Seller $\rightarrow$ Writer, short position.

Price of Option = premium

"On the day" $\Rightarrow$ European Option.

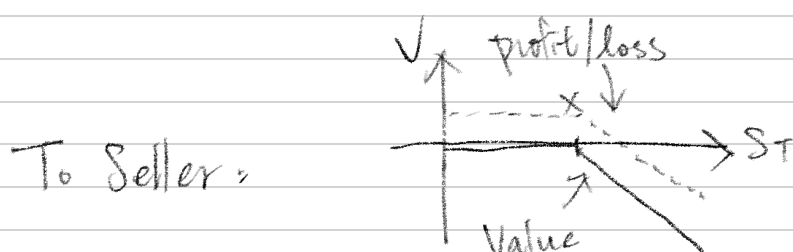
on / before expiring day $\Rightarrow$ American Option.



$$V_T = \begin{cases} 0 & S_T < X \\ S_T - X & S_T > X \end{cases} = \max \{ S_T - X, 0 \}$$

Profit = $\max \{ S_T - X, 0 \} - C_0$

Annotations: $\uparrow$ profit. $\uparrow$ loss (premium) or break even.



$$V_T = -\max \{ S_T - X, 0 \}$$

$$\text{Profit} = C_0 - \max \{ S_T - X, 0 \}$$

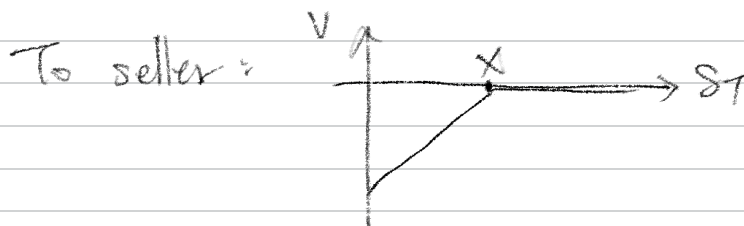
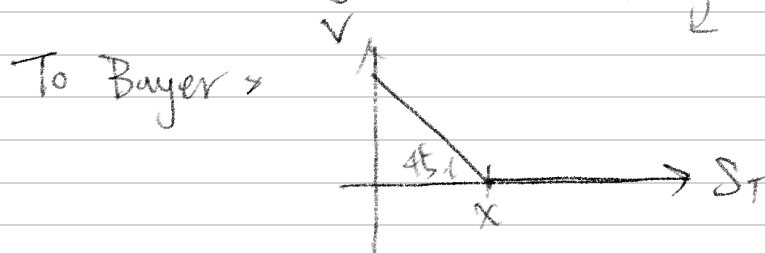
P.s. buy Call if you think stock $\uparrow$

Sell $\sim$ stock $\downarrow$

$S_t = X \Rightarrow$ at the money

$S_t < X \Rightarrow$ out of the money $S_t > X \Rightarrow$ in the money

- Put Option = gives buyer right, not obligation, to sell an asset at a predetermined price on/before date of expiry (assuming European)

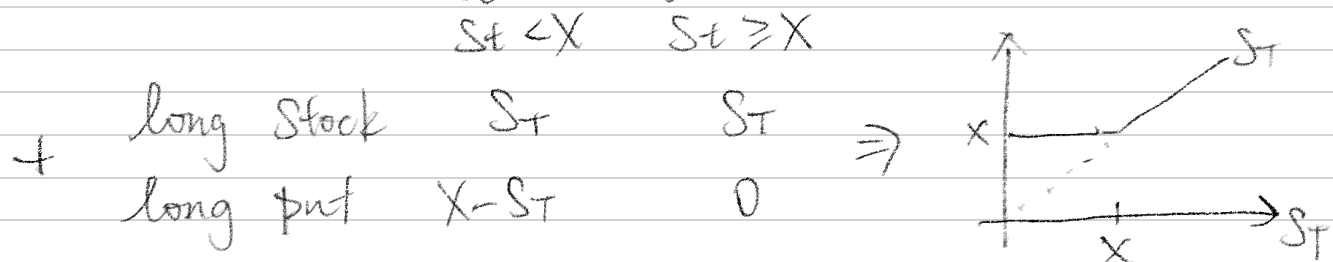


Bullish: strategy non-decreasing w/ stock price

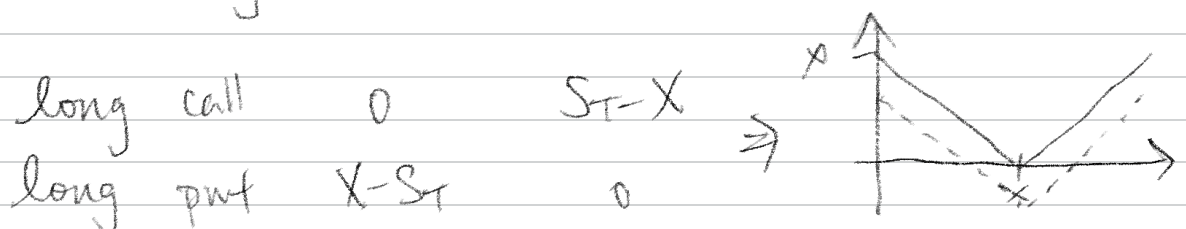
Bearish: ... non-increasing ...

Protective call strategy = buy r_f and call option

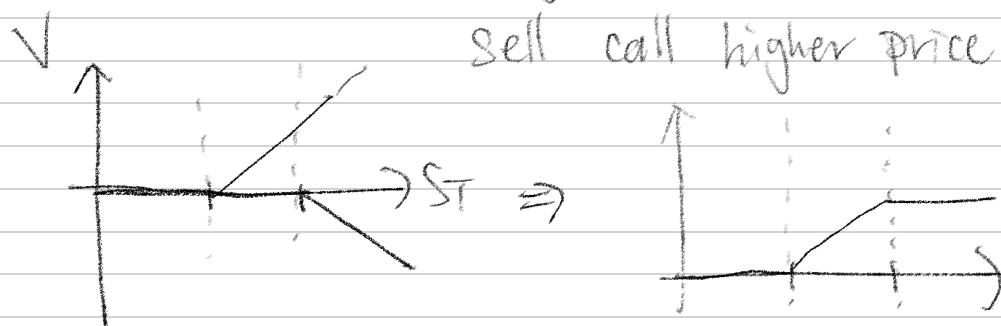
Protective put strategy = buy stock and put at X



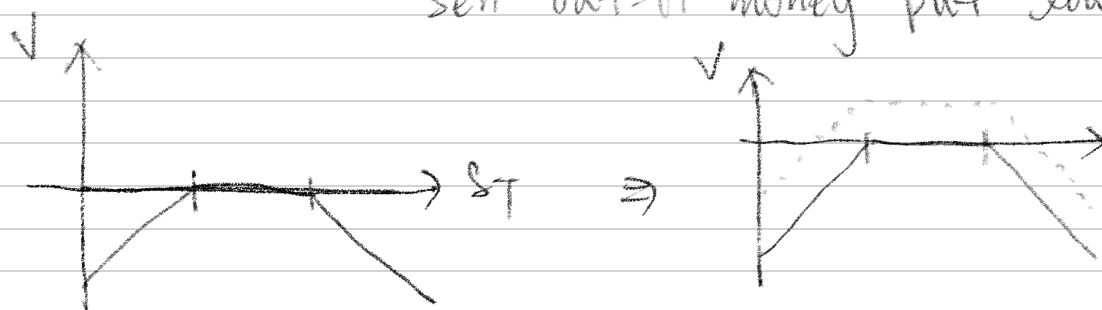
Straddle = long call & Put.



Bull call spread = Buy call lower price,
sell call higher price.



Short-Strangle: sell out-of-the-money call higher price
sell out-of-the-money put lower price.



Option use case:

Put: when you own stock, for protection.

--- don't own stock, strategies.

Call: usually don't own stock.

sell call when you own stock (for premium)

Equity is like call option b/c.

- $V = D + E$

- debt holder get paid first, equity holder last

$$E = \max\{0, V - D\} \rightarrow \text{like call w/ } X = D$$

Put-Call parity:

C = price of call, P = price of put.

$$C + \frac{X}{(1+r_f)^T} = S_0 + P \quad (\text{discrete})$$

$$\text{or } C + \underbrace{X e^{-r_f T}}_{\text{PV of strike price}} = S_0 + P \quad (\text{cts})$$

$$\Rightarrow \text{Call} + \text{bond} = \text{stock} + \text{Put}$$

Binomial Pricing theorem.

One Stage:

$$S_0 \begin{cases} u S_0 & (\text{up factor}) \\ d S_0 & (\text{down factor}) \end{cases} \rightarrow \text{Stock}$$

$$C \text{ (payoff)} \begin{cases} C_u \\ C_d \end{cases} \rightarrow \text{option}$$

$\hookrightarrow$ could be replicate. (same payoff time T)

- Method 1:

buy Δ unit stock, lend B dollar

$$\text{up/down: } \begin{cases} \Delta u S_0 + B(1+r_f)^T = C_u \\ \Delta d S_0 + B(1+r_f)^T = C_d \end{cases}$$

$\Rightarrow$ solve for Δ & B

$\Rightarrow$ Calculate option price w/ law of one price.

$$\star \Rightarrow C = \Delta S_0 + B$$

- Method 2:

$$\text{hedge ratio } \Delta = \frac{\Delta \text{ option value}}{\Delta \text{ stock value}} = \frac{C_u - C_d}{uS_0 - dS_0}$$

buy Δ unit stock, short/write 1 call

$\Rightarrow$ Calculate combined value for both case

= "risk-free" value X .

$$\Rightarrow \frac{X}{(1+r_f)} = \Delta S_0 - C$$

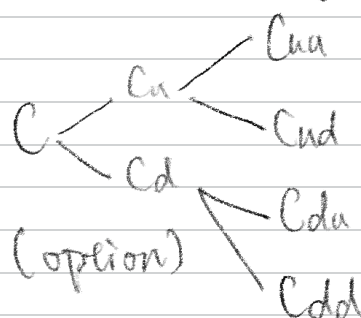
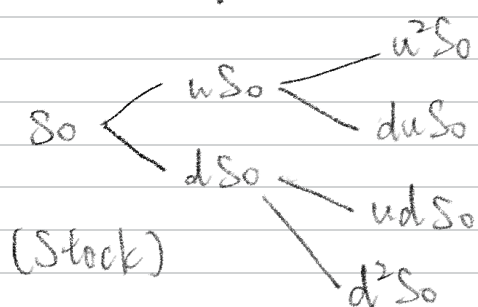
- Method 3:

$$\text{Generally: } C = \frac{1}{1+r_f} (qC_u + (1-q)C_d)$$

$$\text{where } q = \frac{(1+r_f) - d}{u - d}$$

q risk neutral prob. of up move.

Two Stage: (stock price change every 6 month)



Black-Scholes Model

assumption: constant volatility σ^2

stock follow lognormal

$$C = S_0 N(d_1) - X e^{-r_f T} N(d_2) \quad N(0,1)$$

$$d_1 = \frac{\ln(\frac{S}{X}) + (r + \frac{\sigma^2}{2})T}{\sigma \sqrt{T}}, \quad d_2 = d_1 - \sigma \sqrt{T}$$

Exotic Options:

Bermuda option = only exercise on certain date.

blw European & American option.

Look-back-option: payoff depend on extreme price.

$$\text{call payoff} = S_T - \min_{0 \leq t \leq T} S_t$$

$$\text{put payoff} = \max_{0 \leq t \leq T} S_t - S_T$$

Asian option = get average value of stock at expiry.

Forward contract = call + put.

